8.12.2025 Full Marks: 50

Linear Algebra I (MA2102)

End Sem Examination

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Answer all questions. If your total score exceeds 49, you get the full marks. Write your answers as neatly as possible. No points will be awarded for the answers which lack complete justification.

Exercise 1. (4 + 4 + 4 + 4 points)
a) Let A be a 3×3 nondiagonalizable real matrix. Show that all the eigenvalues of A are real.
b) Provide an example of a normal matrix that is neither Hermitian nor unitary.
c) Prove or disprove: If the columns of a matrix A form an orthonormal basis of $\mathbb{R}^n$ w.r.t. the dot product, then the rows of A also forms an orthonormal basis of $\mathbb{R}^n$ w.r.t. the dot product.
d) Let A be an 3×3 real matrix such that $A^{\mathrm{T}} = c_n A^n + c_{n-1} A^{n-1} + \cdots + c_0 I$, for some $n \in \mathbb{N}$ and for some $c_0, c_1, \ldots, c_n \in \mathbb{R}$. Prove or disprove: A is diagonalizable over $\mathbb{C}$.

Exercise 2. (10 + 10 points)
a) Let $\mathbb{P}_2$ denote the space of polynomials with real coefficients and of degree less than or equal to 2. Consider the symmetric bilinear form $\langle , \rangle : \mathbb{P}_2 \times \mathbb{P}_2 \to \mathbb{R}$ given by

$$\langle f, g \rangle = f(-1) \cdot g(-1) + f(0) \cdot g(0) + f(1) \cdot g(1).$$

Find the signature of the form $\langle , \rangle$.
b) Let $T : \mathbb{C}^n \to \mathbb{C}^n$ be a normal operator and let v and v' be two eigenvectors corresponding to two distinct eigenvalues of T. Show that $v \perp v'$ with respect to the standard Hermitian product.

Exercise 3. (4 + 10 points)
a) Let $\mathbb{B} := (v_1, v_2, \ldots, v_n)$ be a basis of a real vector space V and let $\langle , \rangle : V \times V \to \mathbb{R}$ be a symmetric bilinear form with $\langle v_i, v_j \rangle > 0$ for all $i, j \in \{1, 2, \ldots, n\}$. Prove or disprove that $\langle , \rangle$ is a positive definite form on V.
b) Let V be a real vector space and let $\langle , \rangle : V^* \times V^* \to \mathbb{R}$ be an inner product . Show that for every $v \in V$, there exists a unique $f_v \in V*$ such that for all $g \in V^*$, we have $\langle g, f_v \rangle = g(v)$.